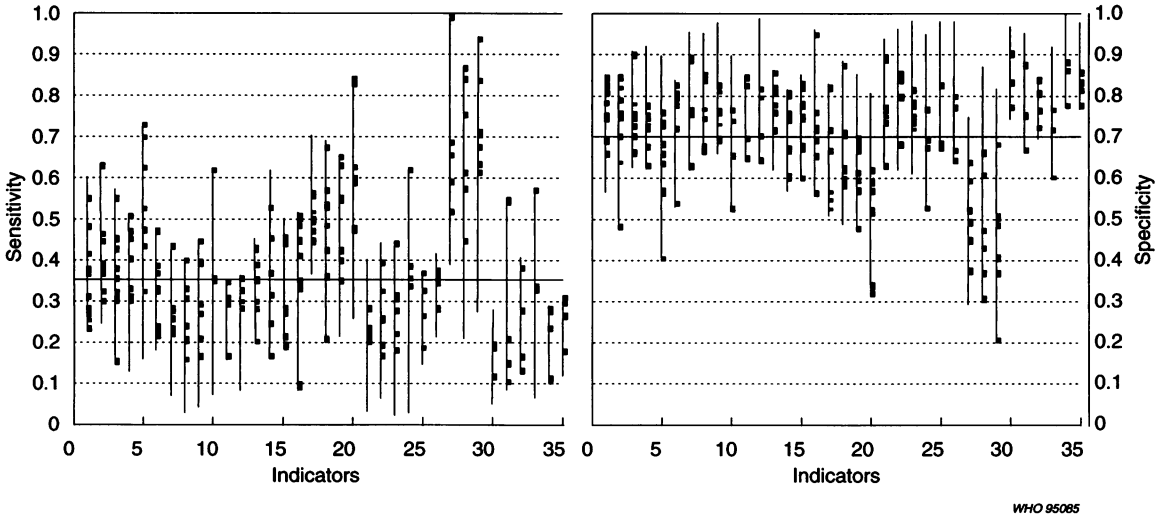
Fig. 7. **Ranges in computed sensitivity (left side) and specificity (right side) across all relevant data sets for all indicators for LBW.** The indicators are numbered 1 to 35 and follow the sequence shown in Table 11.



In summary, the four indicators that meet the full study criteria for consistency are pre-pregnancy weight and attained weight by months 5, 7 and 9 (see Table 11). In this group the maximum Se is 0.56 and minimum, 0.35 (specificity >0.7 in all cases). BMIpp falls just short of the final selection requirements at 38%. The acceptable indicators showed relatively high ORs in Table 10, ranging from 2.3 to 2.5. While ORs for these indicators increased when applied to mothers of low height and low pre-pregnancy weight, the Se/Sp values were not sufficiently consistent across studies to permit their further consideration. The profiles for the marked indicators confirm that these have reasonable reliability across the studies. In service terms, pre-pregnancy or early pregnancy weights are difficult to obtain as registration is often delayed until around the 20th week, particularly in the low-income groups with the highest rates of LBW. This leaves the attained weights at 5 or 7 lunar months as the most useful service indicators for LBW, reflecting current practice in many settings.

Intrauterine growth retardation

Definition: Intrauterine growth retardation (IUGR) has occurred when a birth weight less than the 10th centile of weight-for-gestational age is determined during the analysis by comparing birth weight with reference values derived from a high-income population (*5*). This group includes infants who are full-term (≥37 weeks) growth retarded as well as preterm (<37 weeks) growth retarded. In view of the common 'working definition' of a growth retarded infant, i.e., full-term infant with birth weight less than 2500 g, the analysis was also undertaken for this group of infants.

A summary of the ORs for each of the basic and subgroup indicators, laid out in the same format as for LBW, is presented in Table 12. In the final column, the odds ratios for low-birth-weight infants delivered at full term (FT-LBW) are added.

The biological relationship between maternal nutrition status and IUGR is more precise than for LBW. The papers by Kramer (*4*) and Krasovek & Anderson (*1*) discuss the mechanisms underlying this relationship and also review the past experience of investigators in this area. As stated in the previous section, LBW contains a variable component of preterm births that are not growth retarded but which have low weight at birth. The size of this component will vary under different socioeconomic and health care conditions. Determining the size of the component is often difficult as service operations do not necessarily record a diagnosis of IUGR for lack of suitable reference tables and/or because the appropriate service response does not require such discrimination. For the future, the routine recording of IUGR as a component of LBW rates will greatly enhance the value of service statistics to reproductive epidemiology and subsequent programme planning and evaluation.

ORs for simple indicators range from 1.4 (WTg/7–9) to 3.0 (WT/7 and WT/9). Maternal height and arm circumference show relatively low ORs (1.9 and 1.6) which are similar to those found

Table 12: **Summary of combined odds ratios for each indicator of IUGR.** ORs refer to the relative risk for IUGR for the lowest quartile versus the highest quartile of the indicator's distribution.

Indicator	Odds ratio for combined profiles			
	IUGR	With maternal height[a]	With pre-pregnancy weight[b]	Full-term LBW
Maternal height	1.9			2.0
Mid-upper-arm circumference	1.6			1.9
Pre-pregnancy weight	**2.5**[c]	2.9		2.8
Attained weight by month 5	**2.7**	3.2	3.8	2.7
Attained weight by month 7	**3.0**	3.5	4.0	3.3
Attained weight by month 9	**3.1**	3.4	3.7	3.0
Pre-pregnancy BMI	1.8			2.0
BMI month 5	2.1			1.8
BMI month 7	2.3			2.1
BMI month 9	2.2			1.9
Weight gain:				
Pre-pregnancy to month 5	1.8	2.7	5.4	1.8
Pre-pregnancy to month 7	1.8	2.8	5.2	1.8
Pre-pregnancy to month 9	2.0	3.1	5.5	2.2
Month 5 to month 7	1.7	2.6	2.7	2.0
Month 5 to month 9	1.7	2.6	2.4	1.9
Month 7 to month 9	1.4	2.2	2.6	1.3

[a] In mothers with below average maternal height.
[b] In mothers with below average pre-pregnancy weight.
[c] Highest ORs for the single indicator are given in bold type.

for LBW; these remain virtually the same when applied to full-term LBW. Attained weights at pre-pregnancy and months 5, 7 and 9 show substantially higher ORs which remain similar for full-term LBW. When these indicators are applied to mothers of low stature, they increase by about 0.5 OR units and for low-weight mothers by about 1 unit. BMI shows ORs around 2 which again remain about the same for FT-LBW. Weight gain indicators are all relatively low and are similar for FT-LBW. They increase when modified for low height by about 1 unit and markedly when modified for low weight in respect of gain from pre-pregnancy to months 5, 7 and 9, where they increase by about 2–3 units (although this increase is not evident for the FT-LBW infants). *These latter values are very high (>5) and indicate that low pre-pregnancy weight followed by consistently poor relative weight gain results in a fivefold risk of delivering a growth-retarded infant.* These indicators require measurements relatively late in pregnancy; however, high risk defined by both attained weight and weight gain at 5 lunar months (around 16 to 20 weeks' gestation) provides sufficient time to institute effective intervention where service conditions permit. As with LBW, the reliability of these OR values must be considered and profile plots for the two sets of indicators with high OR values have been selected for this purpose and are presented below.

Weight attained. The combined OR profiles for IUGR by each attained weight indicator are graphed in Fig. 8. The modest progression in the rise in the mean ORs from pre-pregnancy weight through to weight at months 5, 7 and 9 is clear, as is the almost total overlap between the profiles for months 7 and 9, signifying little predictive difference between these. There is also substantial overlap between the profile for pre-pregnancy weight and weight at month 5. This latter indicator shows a broad profile indicative of significant differences in computed ORs for IUGR across the studies.

Weight gain in mothers of below average pre-pregnancy weight. Fig. 9 visually confirms the sharp contrast in performance between the two sets of weight gain indicators for mothers of low pre-pregnancy weight. The overlap between profiles for both sets of indicators is small. The three cross-pregnancy gain

Fig. 8. **The combined OR profiles for attained weight indicators and IUGR are shown.** Means (with 95% confidence interval) are estimated as: WTpp: 2.5 (2.3–2.7); WT/5: 2.7 (2.3–3.2); WT/7: 3.0 (2.7–3.3); and WT/9: 3.1 (2.7–3.4). The ORs are for weight below the lowest quartile versus weight above the highest quartile.

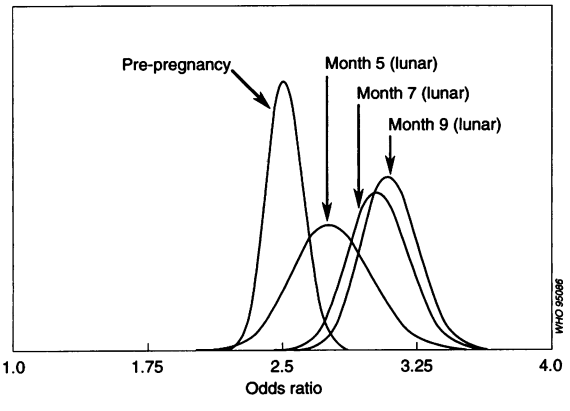
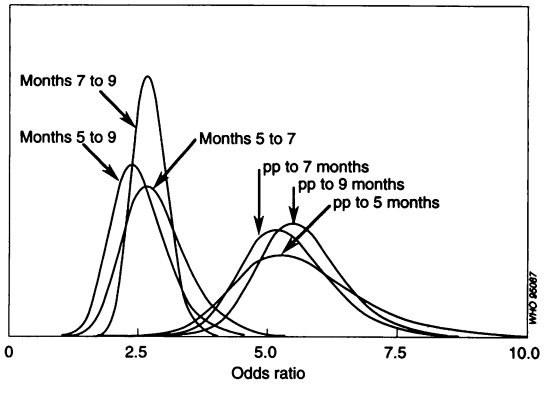


Fig. 9. **The combined OR profiles for weight gain indicators and IUGR are shown.** Means (with 95% confidence interval) are estimated as: WTg/pp–5: 5.4 (3.6–8.2); WTg/pp–7: 5.2 (3.8–7.2); WTg/pp–9: 5.5 (4.1–7.4); WTg/5–7: 2.7 (1.7–4.2); WTg/5–9: 2.4 (1.5–3.7); and WTg/7–9: 2.6 (2.0–3.4). The ORs are for mothers with low weight gain (below the lowest quartile) and low pre-pregnancy weight (below the median) versus weight gain above the highest quartile and above median pre-pregnancy weight.



indicators have OR values around 5.0 indicating very high predictive power, whereas the intra-pregnancy set are centred around 2.5. The ORs for the former set show wide profiles (compared with the intra-pregnancy set) indicating a higher degree of uncertainty in the estimated mean, reflecting both less consistency across studies and the small number of studies with relevant information on which to base the combined estimate.

Sensitivity and specificity

Plots of ranges for sensitivity and specificity across data sets for all indicators for IUGR appear in Fig. 10. The indicators are numbered 1 to 35 and follow the sequence shown in Table 13. Again, the trade-off between Se and Sp at the level of individual indicators will be obvious in these figures. Maximum spread for Se is observed for pre-pregnancy weight (number 2), with values ranging from 0.10 to 0.85. This would normally suggest poor consistency with respect to this parameter across data sets; however, the majority of data sets display a much smaller range from just below 0.3 to just above 0.5. Minimum spread for Se occurs for weight gain from month 5 to month 7 in mothers with low pre-pregnancy weight (number 30), with values of 0.15 to 0.25. But these are unacceptably low and therefore this indicator would not meet the criterion for an acceptable minimum Se with Sp >0.7. A similar conclusion follows for virtually all of the other indicators with high ORs, although a number meet the required Se level but then fail to achieve the minimum specificity.

Table 13 presents the frequency with which a given indicator met the Se/Sp criterion across the studies analysed. "Proportion" indicates the proportion of studies in which this criterion was met and provides a marker of the indicator's usefulness on a regional or global basis. However, it will be noted that this proportion achieves 40% or more for a very few indicators.

Fig. 10. **Ranges in computed sensitivity (left side) and specificity (right side) across all relevant data sets for all indicators for IUGR.** The indicators are numbered 1 to 35 and follow the sequence shown in Table 13.

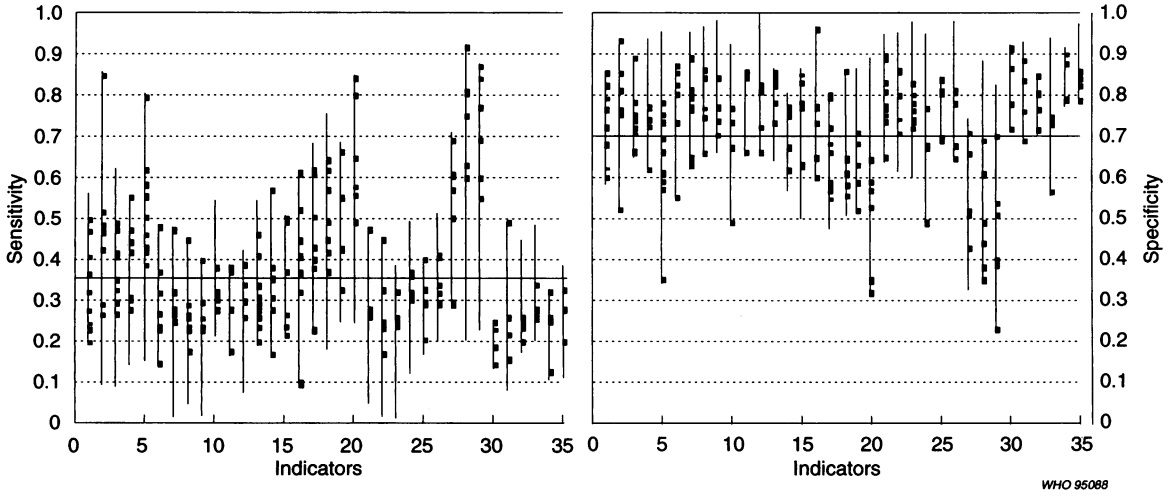


Table 13: **Performance of indicators for IUGR.** The asterisked ones, having met the selection criterion (see text), are considered candidates for 'best' indicator.

Indicator	Intrauterine growth retardation[a]			
	No.	Proportion	Min Se	Max Se
1. Maternal height	23	0.17	0.36	0.41
2. Pre-pregnancy weight	20	0.50*	0.36	0.52
3. Attained weight by month 5	15	0.40*	0.38	0.49
4. Attained weight by month 7	14	0.36	0.36	0.55
5. Attained weight by month 9	17	0.53*	0.36	0.50
6. Mid-upper-arm circumference	13	0.08	0.37	0.37
7. Weight gain: months 5 to 7	12	0.00		
8. Weight gain: months 5 to 9	13	0.00		
9. Weight gain: months 7 to 9	13	0.00		
10. Weight gain: pp to month 5	8	0.00		
11. Weight gain: pp to month 7	8	0.13	0.38	0.38
12. Weight gain: pp to month 9	8	0.25	0.39	0.42
13. Pre-pregnancy BMI	20	0.30	0.38	0.46
14. BMI month 5	14	0.21	0.36	0.41
15. BMI month 7	13	0.23	0.36	0.41
16. BMI month 9	15	0.27	0.36	0.45
In mothers with low maternal height:				
17. Pre-pregnancy weight	18	0.06	0.60	0.60
18. Attained weight by month 5	13	0.31	0.37	0.47
19. Attained weight by month 7	13	0.15	0.42	0.47
20. Attained weight by month 9	13	0.15	0.50	0.52
21. Weight gain: months 5 to 7	11	0.09	0.36	0.36
22. Weight gain: months 5 to 9	12	0.08	0.45	0.45
23. Weight gain: months 7 to 9	12	0.17	0.36	0.38
24. Weight gain: pp to month 5	8	0.00		
25. Weight gain: pp to month 7	8	0.13	0.38	0.38
26. Weight gain: pp to month 9	8	0.13	0.51	0.51
In mothers with low pre-pregnancy weight:				
27. Attained weight by month 5	8	0.00		
28. Attained weight by month 7	8	0.13	0.51	0.51
29. Attained weight by month 9	8	0.00		
30. Weight gain: months 5 to 7	6	0.00		
31. Weight gain: months 5 to 9	6	0.00		
32. Weight gain: months 7 to 9	7	0.14	0.44	0.44
33. Weight gain: pp to month 5	8	0.00		
34. Weight gain: pp to month 7	8	0.00		
35. Weight gain: pp to month 9	8	0.13	0.39	0.39

[a] No. is the number of data sets for which the indicator could be estimated. Proportion gives the proportion of data sets in which the Se/Sp criterion was met. The asterisk denotes that the indicator met the criterion (Sp >0.7 and Se >0.35) in over 40% of the studies. The observed maximum and minimum sensitivities across studies that meet the criteria are recorded in the final two columns.

In summary, it can be seen that pre-pregnancy weight and attained weight by months 5 and 9 are acceptable indicators on the basis of cross-study consistency (Table 13). They have minimum and maximum sensitivity from 0.36 to around 0.5 (with specificity >0.7). It will be recalled that ORs for these indicators range from 2.5 to 3.0, which represents a relatively strong predictive capacity. In the same indicator category, attained weight by month 7 is just short of the full test criterion at 36%. Clearly, in cases where fifth month information was unobtainable this would be a reasonable substitute carrying a high OR of 3.0, rising to 3.5 and to 4.0 in mothers of below average maternal height and maternal weight respectively.

Preterm birth

Definition. Preterm birth occurs when the gestational age at delivery was less than 37 weeks. A summary for odds ratios is presented in Table 14. Indicators measured in the 9th lunar month have not been included as the great majority of preterm births had occurred at this stage (<37 weeks).

There is no obvious biological route through which maternal nutritional status may affect gestational age at delivery or initiate preterm birth (PTB). It will be recalled, however, that maternal anthropometric indicators are analysed in this study for their predictive power, which does not necessarily imply a causal relationship. Kramer's analysis of the literature does include an examination of anthropometric indicators in connection with both gestational age and prematurity (*4*). That study did not find any statistical relationship and he noted that the majority of premature births remained unexplained.

Table 14: **Summary of the combined odds ratios for each indicator.** ORs refer to the relative risk for preterm birth for the lower quartile versus the upper quartile of the indicator's distribution.

	Odds ratio for combined profiles		
Indicator	Preterm birth	With maternal height[a]	With pre-pregnancy weight[b]
Maternal height	1.2		
Mid-upper-arm circumference	1.2		
Pre-pregnancy weight	**1.4**[c]	1.4	
Attained weight by month 5	0.9	1.0	0.9
Attained weight by month 7	0.9	0.9	1.0
Pre-pregnancy BMI	**1.3**[c]		
BMI month 5	0.7		
BMI month 7	0.9		
Weight gain:			
Pre-pregnancy to month 5	0.4		
Pre-pregnancy to month 7	0.7		
Month 5 to month 7	**1.4**[c]	1.8	1.6

[a] In mothers with below average maternal height.
[b] In mothers with below average pre-pregnancy weight.
[c] Highest ORs for the single indicator are given in bold type.

The range of ORs is from <1 to a maximum of 1.8 (WTg/5–7 in mothers of below average height) and it is evident that none of these indicators are strongly predictive of a risk for this outcome. Attained weight at months 5 and 7 and BMI at month 7 have ORs around 1.0. This implies that there is no evidence that risk to a mother of low weight at these stages of pregnancy differs from a mother of high attained weight. Maternal height, arm circumference, pre-pregnancy BMI, and weight gain between months 5 to 7 all have estimated mean ORs exceeding 1.0 by a relatively small margin. The 95% confidence intervals for these indicators are as follows (in the same order): 1.1–1.2, 1.0–1.3, 1.3–1.5, 1.1–1.4, and 1.1–1.7. Except for MUAC, the lower interval value in each case is 1.1 or higher; this suggests that the effect is real, but remains too small to be of practical importance. Of the remaining general indicators, three actually have estimated confidence intervals wholly below 1.0: BMI at month 5 (0.6–0.9), weight gain from pre-pregnancy to month 5 (0.3–0.6), and weight gain from pre-pregnancy to month 7 (0.6–0.9). The stated objective of this project is to establish useful indicators of risk in the sense of preventing disease and disability. It is nevertheless interesting to consider evidence for reduced risk. For this outcome the evidence suggests that low weight gain from pre-pregnancy to month 5, as compared to a high gain during this period, reduces the relative risk of a preterm delivery to about 40%. The evidence is inconsistent however, as the next indicator (gain for months 5 to 7), which represents the same mothers at a different stage of pregnancy, indicates a positive risk (OR, 1.4) which increases when subgroups with low maternal height (1.8) and low pre-pregnancy weight (1.6) are considered. The OR profiles for the general indicator sets containing the highest mean values are described below.

Maternal pre-pregnancy weight and weight attained at month 5 and 7. The plot of the combined studies profile for pre-pregnancy is well separated from the other attained weight indicators (months 5 and 7) and has a comparatively narrow spread indicating good reliability (Fig. 11). *Low pre-pregnancy or early pregnancy weight therefore increases the relative odds of a preterm delivery slightly.*

Body mass index. Within the BMI group, only pre-pregnancy BMI had an OR exceeding 1 (Fig. 12).